Zain Dana Harper

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Curriculum Vitae | Systems Engineering, AI Infrastructure, and Field Operations

Profile

Self-directed software builder and technical writer with about 13 years of personal programming. Builds AI workstations, evaluation tools, language tooling, and graphics systems. Began a career transition in June 2026 after eleven years of full-time production arboriculture; no professional programming experience.

Open to technical and nontechnical AI governance and evaluation roles. Willing to relocate to London; available for regular travel to San Francisco.

Self-Directed Projects

About 13 years of self-directed programming | Career transition began June 2026

Sole proprietor since September 2026; no paid software clients

- Own the architecture, implementation, integration, tests, documentation, and release work for Flywheel and related tools. Work spans Python services and command-line tools, Flutter desktop software, Rust systems tooling, and browser interfaces.
- Develop model-agnostic workflows, evaluation harnesses, controlled-action interfaces, and source-linked records that can be checked without relying on a model to approve its own answer.
- Contribute focused fixes, regression tests, and documentation to externally maintained projects, responding to maintainer review and preserving the scope of each accepted change.

Freelance Technical Writer, Documentation, and Product Operations

Independent practice | Began 2017; project-based work

- Produced technical and compliance documentation, API and implementation guides, proposals, release notes, and operational playbooks. Turned implementation details into material usable by customers, developers, and reviewers.

Employment

Commercial Arboriculture and Field Operations | Legendary Tree

Seattle area | April 25, 2015 to June 2, 2026

- Combined ground-based tree-care work and safety coordination with client intake, site assessments, estimates, proposals, scheduling, and crew/vendor communication.
- Monitored rigging systems, clearances, and changing conditions while supporting climbers from the ground. Explained work options and constraints to clients and maintained continuity between the proposed scope and field execution.

Technical Networking Support | Xbox/Microsoft contract

Subcontracted through Stream/Convergys | Wilsonville, Oregon | 2014 to 2015

- Diagnosed connectivity and account issues by phone and chat using TCP/IP, DNS, NAT, firewall, and router troubleshooting; documented repeatable resolutions and customer-facing instructions.

Programming Background

Personal programming began with game modding, reverse engineering, bot and rotation logic, and runtime-memory tooling, then developed into systems engineering, graphics, compilers, and AI infrastructure.

AI Platform and Agent Systems

Flywheel | Model-agnostic AI workstation and coding harness

Primary owned platform. Flywheel v1.9.0, released October 15, 2026, is published on PyPI as flywheel-verify, installs through Homebrew or Scoop, and ships a Windows installer with a bundled engine. The source repository contains the Python runtime, Flutter desktop client, and browser fallback.

- Routes tasks to hosted or local models through an OpenAI-compatible gateway; keeps model selection separate from the task record and exposes command-line, desktop, and API entry points.
- Provides Rowan, a desktop assistant that translates requests into recorded tasks, together with permission-checked coding and tool-use workflows, session state, and inspectable outcomes.
- Composes research intake, workspace indexing, verification, orchestration, memory, authoring, and other tools through registered lanes and published interfaces. Bundled components and installable companions have distinct packaging requirements.
- Produces sealed, re-derivable receipts and offline MATCH, DRIFT, or UNVERIFIABLE outcomes. A reproducible receipt establishes the bounded check it records, not universal model accuracy or safety.
- Released a monitor-admission check over 2,676 frozen items with six planted controls, a second independent math kernel, and signer-head anchoring. Release evidence reports no real monitor tested through the admission check and no installed signer; Windows anchoring was parse-checked only.
- Added behavior-check in v1.9.0: it replays 13 scripted sessions on two builds and names the first event that differs. On a corpus written before it ran, it flagged 20 of 20 planted behavior changes and 0 of 50 unchanged pairs. These checks used a scripted model on Windows; no hosted model was called.

Accountable Surface | Governed actions and computer-use adapters

Built operator-granted action interfaces for filesystem, command, API, browser, and Windows UI Automation work. Actions carry explicit scope and postconditions; persisted journals preserve decisions and observed effects. Reversible adapters include restoration paths. Fake drivers make browser and desktop scenarios testable without uncontrolled live actions.

The v0.2.0 release provides source and wheel artifacts on GitHub. The project remains alpha; browser integrations are optional and a tool refusal or unknown result is not promoted into a successful action.

Articulate | Writing analysis and accountable editing

Built deterministic prose analysis with profiles for register and genre, mode settings, and optional model-assisted editing. Includes command-line, library, MCP, language-server, and SARIF surfaces; math-span protection preserves formulas during supported editing flows. Content-free audit receipts record checks without retaining the underlying prose. Version 0.14.0, released October 23, 2026 and published on PyPI as articulate-writing, masks Markdown structure before the detector rules run. On 3,000 Markdown files the masks removed 5,123 findings on table separator rows and added none; the detector rules and their fingerprint did not change.

Daylight Commons | Browser extension for checkable evidence

Built a browser extension that shows the sourced evidence behind replies on X, Reddit, YouTube, Bluesky, Mastodon, and other sites. Each sign names its source and states what it does not prove; the extension never labels an account a bot and runs with no server, account, or analytics. Version 3.0.0, released October 23, 2026 under AGPL-3.0, stores evidence lookups as a hash-chained file that a command-line tool verifies offline. No Mozilla-signed build exists yet, and most signs have no measured precision.

Agent tools, memory, and coordination

Relay supplies model-agnostic coding-agent tooling; Plexus connects tool interfaces; Mneme retains memories with source checks; Canon consolidates assistant instructions and context. Bulletin provides a signed-identity public message board. These systems contribute distinct capabilities to the larger platform. Customer-scale deployment is not established.

Evaluation, Systems, and Security Engineering

Terminal State Fixtures and Crucible

Built 324 exhaustive terminal-state fixtures and five synthetic component-evaluation families with 337 cases and 5,058 frozen expectations. Defined controls separate failed outputs, provider failures, and missing evidence from valid results. Executed a pinned METR count_odds interoperability check with three deterministic controls, preserving logs and score provenance through the Inspect-to-Flywheel handoff.

EMET | Byte-integrity witnessing

Developed Python, Rust, Node, and Go implementations around shared conformance cases and portable verification records. The witness checks byte-level integrity, anchors, and source correspondence; it does not grant permission, certify semantic truth, or replace an operator decision. Cross-language checking makes implementation differences visible.

BuildLang | Rust compiler and developer tooling

Developed a systems-language toolchain with typed capability effects, type inference, parser and intermediate-representation work, native C output, shader output, and language-server support. The compiler makes categories of ambient access visible to its type and build machinery. Version 1.8.0, published to crates.io on October 23, 2026, adds Box<T> and recursive types. For 1.7.0, 27 held-out tasks were sealed by hash first; two fresh agents scored 20 and 17 of 27, short of the 22 set in advance, and each failure class became an issue.

Phantom | Hardware-identity inspection and reversible system changes

Developed a Rust-based Windows and Linux utility with deterministic profiles, read-only audit, backup-before-write behavior, validation, restoration, and platform packages. The released v1.1.1 scope is userland-reversible identifiers. Kernel and firmware layers are modeled, not shipped.

Controlled assessment and operator tooling

Private systems work includes Array for approval-gated assessment orchestration and containment records, Lab/Seed for controlled-lab fixtures and detection-engineering workflows, and Runtime/ORCA for local workspaces, reports, bundles, and release provenance. Public descriptions cover architecture and safe interfaces; private implementations, target material, and operational details are not published here.

Research intake and workspace tooling

Built Gather for source-carrying research intake, Index for workspace and symbol maps, and Forum for model orchestration with recorded decisions. These tools make source selection, handoffs, and task context inspectable across a multi-repository product. On October 23, 2026, Gather 2.4.0, Crucible 1.6.0, and EMET 1.3.1 shipped to PyPI and Canon 0.7.0 to GitHub, each documented with a checked first run, a comparison with adjacent tools, and stated limits.

Graphics, Color, and Runtime Work

Developed HLSL and C++ graphics and game-modding work spanning post-processing, Direct3D 11 integration, shader debugging, GPU traces, and shared-memory interfaces. Related public projects include Elder ENB, SkyrimBridge, ENB Runtime Core, Brender Archival, and Engine Revival. Existing engines and collaborators retain their own attribution.

raw-native is a C++ renderer that draws each scene on a CPU reference and on the GPU through D3D12 or WebGPU, then writes a certificate stating whether the two agree within committed tolerances. Version 0.8.0, released October 14, 2026, produced byte-identical default renders from its Windows, D3D12, Linux, and WebAssembly builds and passed 60 of 60 tests with MSVC and GCC. No hardware-GPU render was checked for that release; Vulkan is an optional hardware probe, not a render backend.

Color and creative-tool work includes Build Color, Calibrate Pro, Studio Engine, and browser-based studio/print surfaces: perceptual color spaces, color difference, HDR/tone-mapping tools, display characterization, and generative visual systems. These are engineering projects, not claims of professional calibration certification.

Selected Open-Source Contributions

Six selected contributions merged across six external repositories, covering bug fixes, tests, and documentation. These are contributions, not employment; the broader record tracks merged, open, and closed without merge outcomes.

- DeepEval #2822: fixed unhashable tool-output and nested-parameter failures in tool-correctness evaluation; merged July 2, 2026. github.com/confident-ai/deepeval/pull/2822
- TOMLKit #549: preserved dotted-key siblings when replacing a value with an array of tables; merged July 14, 2026. github.com/python-poetry/tomlkit/pull/549
- Datasette #2815: corrected malformed composite-key requests from HTTP 500 or incorrect HTTP 200 to HTTP 400; tested too few and too many key components in HTML and JSON. Merged July 7, 2026.

github.com/simonw/datasette/pull/2815
- Free Law Project #820: documented runnable database-free fast tests; merged August 25, 2026. github.com/freelawproject/litigant-portal/pull/820
- Hebbian Robotics #157: tested reserved-column drift against the actual queryable episode schema; merged August 25, 2026. github.com/Hebbian-Robotics/hflow/pull/157
- Voxwire #44: added offline regression coverage for a six-event degraded voice turn, matching saved PCM audio and JSONL records to emitted events. Merged August 15, 2026. github.com/vectorvoyager358/voxwire/pull/44

Technical Competencies

Languages and runtimes: Python, Rust, C++, Dart/Flutter, TypeScript/JavaScript, Lua, HLSL, C#, PowerShell, and Bash. Additional cross-language conformance work includes Go and Node implementations.

AI infrastructure and quality: multi-provider/local-model routing, tool-use loops, task decomposition, MCP, evaluation design, deterministic fixtures, regression testing, worker/verifier separation, failure analysis, and terminal-state scoring.

Verification and controlled actions: capability effects, operator grants, action envelopes, source provenance, hash-chained journals, offline replay, scoped postconditions, conformance suites, and release evidence.

Software delivery: Git/GitHub, GitHub Actions, Python packaging, command-line design, Windows/Linux installers, CMake, test automation, static websites, release documentation, and checksums.

Graphics and domain knowledge: D3D11/D3D12, HLSL, WebGPU/WGSL, CPU reference rendering and GPU parity checks, shader and runtime debugging, color-space and tone-mapping work, GPU instrumentation, shared-memory interfaces, and applied arboriculture/horticulture from field practice.

Working Practice

AI-assisted development is part of the workflow. Architecture, integration, source review, test design, release decisions, and responsibility for the submitted work remain mine. Public examples separate observed results from interpretation and retain negative or inconclusive findings. A completed run alone does not establish success.

Research Agenda

My independent research examines evidence that another person or system can check without inheriting an AI system's authority. Work includes evaluation provenance and verifier independence; live-state drift; proof-carrying actions; memory, uncertainty, and behavior; and human authorization for consequential actions. Robotics is a research interest.

AI Accountability, Disclosure, and Public-Interest Writing

Independent essays on AI accountability, disclosure, conflicts of interest, and human consequences, relevant to public-interest and whistleblower-protection work.

Who Knew First

September 2026. Directed and published an AI-assisted investigation comparing disclosure timelines, evidence access, institutional incentives, and evaluator conflicts across nine reported AI-agent incidents. Separates source claims, inference, and unresolved attribution. harperz9.github.io/who-knew-first.html

An Open Letter on Checking the Machines

September 2026. Proposed contestable AI decisions, repeatable evidence checks, and accountable responses to objections. harperz9.github.io/checking-the-machines.html

Borrowed Ground

September 2026. Philosophical essay connecting human interdependence to responsibility, privacy, participation, and the limits of technical evidence as authority over people. Substantial AI assistance disclosed. harperz9.github.io/borrowed-ground.html

The Sandbox Was Never Just a Box

September 2026. Compared public security reports to examine shared-service exposure and the independence of logs, monitors, and scorers in AI-agent evaluation. harperz9.github.io/the-sandbox-was-never-just-a-box.html

Selected Publications and Research Records

Independent systems papers, preprints, research notes, philosophy papers, a working paper, and archived manuscripts with DOI records. These records are not peer reviewed. An archived philosophical manuscript is not an awarded academic degree.

- EMET: An Authority-Incapable Byte-Level Integrity Witness. Systems paper, 2026. Byte-integrity witnessing and the separation of observation from authority. doi.org/10.5281/zenodo.21230267
- BuildLang: Accountable Compute via Typed Capability Effects. Systems paper, 2026. Capability effects and accountable compilation. doi.org/10.5281/zenodo.21231253
- Witnessed Independence: Recording Whether a Verifier Graded Its Own Work. Preprint, 2026. doi.org/10.5281/zenodo.21232206
- Proof Packets: A Derive-Don't-Trust Envelope for Accountable Agent Actions. Preprint, 2026. doi.org/10.5281/zenodo.21231406
- The Personhood-Gate Handoff. Research note, 2026. doi.org/10.5281/zenodo.21234475
- Re-Perceived Effects. Research note, 2026. doi.org/10.5281/zenodo.21231311
- Faithfulness to a Named Criterion Is the Conserved Quantity Across Lossy Substrates. Research note, 2026. doi.org/10.5281/zenodo.22768398
- The Witnessing Spine. Archived research corpus, 2026. doi.org/10.5281/zenodo.20778927
- Conferred Existence. Long-form philosophical manuscript and archived corpus, 2026. doi.org/10.5281/zenodo.20773724
- Sourcehood Without a Source: The Arity Gap and the Free Will Debate. Philosophy paper, edition of 3 October 2026. doi.org/10.5281/zenodo.23126117
- The Seam That Holds While It Is Crossed. Philosophy paper, edition of 3 October 2026. doi.org/10.5281/zenodo.23126357
- Self-Given Is Not Self-Grounding. Philosophy paper, edition of 3 October 2026. doi.org/10.5281/zenodo.23126458
- The Conservation of Faithfulness. Working paper, edition of 3 October 2026. doi.org/10.5281/zenodo.23126484

Education

High School Diploma | Wilsonville High School, Wilsonville, Oregon | 2013

Self-directed study in systems programming, compilers, graphics, color science, AI evaluation, epistemics, and philosophy of mind. Practical arboriculture and plant-science knowledge comes from field work; no separate professional license is claimed.

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Publications: harperz9.github.io/publications.html | Portfolio: harperz9.github.io/portfolio.html

Credentials and Continuing Education

2026 completions. Verification: harperz9.github.io/career/source-ledger.md.

Qualys Certified Specialist: Policy Audit; Custom Assessment and Remediation; Vulnerability Management Detection and Response; Qualys Kubernetes and Container Security; Qualys TotalCloud Exam; Endpoint Detection and Response | Issued 2026-09-30; expire 2028-09-29

Microsoft Applied Skills | Earned September 2026

- Secure AI solutions in the cloud
- Get started with identities and access using Microsoft Entra
- Secure storage for Azure Files and Azure Blob Storage
- Implement security through a pipeline using Azure DevOps | Intermediate; 95% assessment

Advanced Microsoft Learn Coursework

SC-100: Design security solutions for applications and data. Completed four module assessments and the architecture case study.

Design solutions that align with MCRA and MCSB. Advanced Microsoft Learn module completed September 30, 2026: security architecture, Zero Trust, insider and supply-chain risk, and AI security controls.

Linux Foundation Course-Completion Certificates

LFEL1012: Secure AI/ML-Driven Software Development; LFS120: Conversational AI: Ensuring Compliance and Mitigating Risks; LFD121: Developing Secure Software

LFS182: Securing Your Software Supply Chain with Sigstore; LFEL1001: Understanding the EU Cyber Resilience Act (CRA); LFS118: Ethical Principles for Conversational AI

LFEL1006: Securing Projects with OpenSSF Scorecard; LFEL1005: Security Self-Assessments for Open Source Projects; LFEL1007: Automating Supply Chain Security: SBOMs and Signatures

LFD125: Security for Software Development Managers; LFS180: Introduction to DevSecOps for Managers; SKF100: OWASP Top 10 Security Threats

LFS184: Introduction to JavaScript Security; LFC108: Cybersecurity Essentials; LFEL1004: Authentication and Authorization for Web/API

LFEL1008: Migrating AI Workflows Across Platforms; LFS147: Introduction to AI/ML Toolkits with Kubeflow

Additional Coursework and Skill Badges

Cisco Networking Academy: Ethical Hacker | Intermediate course certificate and learning badge, 2026-09-30; 100% knowledge exam. Assessed scoping, reporting, vulnerability assessment, and security concepts.

Google skill badges, completed 2026-09-30: Google DeepMind: Train A Small Language Model (Advanced; 100/100, four coding checkpoints); Use Agent Skills with Multi-Agent Systems (Intermediate course/Advanced challenge; 100/100, six checkpoints). Scope: tokenization, n-grams, data preparation, and agent orchestration. Multi-agent lab: five Cloud Run services deployed; model-access policy blocked local and cloud generation.

Google BigQuery skill badges, issued 2026-10-01: Perform Predictive Data Analysis in BigQuery (100/100, seven checkpoints; Advanced challenge); Create ML Models with BigQuery ML (100/100, four checkpoints; Intermediate challenge). Both courses Intermediate. Lab work: SQL features, logistic regression, later-window evaluation, and predictions.

Google course-completion badges, completed 2026-09-30: Machine Learning Operations (MLOps) with Agent Platform: Model Evaluation (Intermediate); Google Security Operations - Deep Dive (Advanced; 96% knowledge assessment). Topics: model-evaluation methods, YARA-L, SIEM processing/enrichment, and SOAR.

Anthropic Claude Academy completion badges: Model Context Protocol: Advanced topics; Claude Code in action; Building with the Claude API; Claude with Amazon Bedrock | Issued September 30, 2026

UNESCO: Global MOOC on the Ethics of AI | Intermediate course certificate, 2026-09-30; 100%. Assessed human rights, fairness, risk controls, incident reporting, and sustainability.

BlueDot Impact: AGI Strategy | Course completed in 2026

Developed an AGI Strategy action plan for label-independent AI incident notice, independently reviewable evidence, accountable escalation, and tests for misleading verification results.

Coursework: MCRA/MCSB, Zero Trust, OWASP web Top 10, CRA. Research references: NIST AI RMF/GenAI Profile, ISO/IEC 42001/23894, ISO 37002. Scope: harperz9.github.io/career/standards-reference.md.